

LE NGUYEN DUY LUAN

Mobile Developer | On-device AI / Computer Vision

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Detail-oriented Mobile Developer specializing in Native Android (Kotlin, Jetpack Compose) and On-device AI/Computer Vision (TensorFlow Lite). Experienced in building performance-optimized, offline-first mobile applications with real-time ML capabilities and cloud-backed sync (Firebase, Supabase). Strong problem-solving skills and dedication to writing clean, maintainable code.

EDUCATION

Nha Trang University

Bachelor of Information Technology

Graduated: Mar 2026

Nha Trang, Vietnam

- **GPA:** 7.94/10 (3.25/4.0)
- **Graduation Project:** Lung X-ray Disease Classification using Segmentation-guided Masking — Score: 9.1/10
- **Relevant Coursework:** Data Structures & Algorithms, Mobile Device Programming, Software Architecture & Design, Image Processing, Artificial Intelligence

TECHNICAL SKILLS

Languages: Java, Kotlin, JavaScript, Python, SQL

Mobile: Android SDK, Jetpack Compose, CameraX, Room DB, WorkManager, Coroutines & Flow

AI / Computer Vision: TensorFlow Lite, MediaPipe, MLKit (Pose Estimation), YOLO, OpenCV, Gemini API

Backend Services: Firebase (Auth, Firestore, Storage), Supabase, REST API design, OkHttp

Tools: Git/GitHub, Android Studio, Gradle, ADB (Android Debug Bridge), KSP, VS Code, Figma (basic)

AI-Assisted Dev: Google Antigravity, OpenAI Codex

Languages (Spoken): Vietnamese (native), English (technical reading/writing)

PROJECTS

VocabLensAI — Native Android English Learning Platform | *Personal Project*

- **Tech Stack:** Kotlin, Jetpack Compose, Coroutines/Flow, Room DB, CameraX, TensorFlow Lite, Gemini API, Firebase, Supabase.
- Developed an offline-first Android application from scratch using Kotlin, Jetpack Compose, MVVM, and clean code practices.
- Solved the problem of passive vocabulary memorization by implementing real-time on-device object detection (CameraX + TensorFlow Lite), integrating Gemini AI to generate contextual learning stories from scanned vocabulary.
- Addressed the “forgetting curve” problem in language learning by designing a custom SM-2 Spaced Repetition engine that automatically schedules reviews at optimal intervals, backed by a hybrid Room DB/Firestore sync system to ensure reliable offline data sync.
- Built a real-time multiplayer PvP Arena using Firebase Firestore with an ELO rating calculator ($K = 40/20$), alongside a Chess.com-style game review UI for match accuracy analysis and AI feedback.

Lung X-ray Disease Classification | *Graduation Project, Nha Trang University*

- **Tech Stack:** Python, TensorFlow/Keras, OpenCV, U-Net (Segmentation), ResNet/DenseNet (Classification).
- Solved the classification noise problem by developing a U-Net segmentation pipeline to isolate lung regions in chest X-ray images, boosting diagnostic accuracy for ResNet/DenseNet classifiers (Score: 9.1/10).
- Documented the end-to-end medical imaging pipeline from pre-processing to Grad-CAM heatmap visualization, providing interpretable model outputs for academic evaluation.

AI Fitness Coach (Android) | *Personal Project*

- **Tech Stack:** Kotlin, Jetpack Compose, MediaPipe/MLKit (Pose Estimation), Room DB, CameraX.
- Developed an offline-first Android app (Kotlin, Compose, MVVM) combining personalized workout planning with real-time form feedback.
- Addressed the injury risk of unsupervised home workouts by implementing camera-based pose estimation (MediaPipe/MLKit) to extract joint coordinates and calculate angles in real-time.
- Built a custom rule-based feedback engine (e.g., “Keep your back straight”) and automated rep counting, caching history 100% offline via Room DB.